

Q1. Power systems optimisation & algorithms

- (a) Give an overview of the three main types of power allocation problems in power systems. Note their uses in industry, advantages and disadvantages.

1. Economic dispatch

- does not consider power flows or constraints
(sometimes penalty factors are used to approximate this)
- seeks to select the cheapest generators to run
- used in markets to determine which generators to run.
see lecture notes for a formulation & example.

2. DC optimal power-flow

- considers a simplified version of the power-flow equations (linearized, assuming no resistance, reactive power, etc.)
- used to dispatch generators every 5-30 min
- solution is checked for feasibility in AC power-flow

3. AC optimal power-flow

- consider the full AC power-flow equations as constraints.
- computationally very expensive, especially if security constraints are considered, so not used often.

	Use in industry	Computational cost	Accuracy of model
ED	Markets	Very low	Low
DC OPF	Dispatch	Moderate	Moderate
AC OPF	Little	High	Good

(b) Explain what a "penalty factor" in economic dispatch is.

A penalty factor is used to weight solutions towards local generation, thus reducing power losses & congestion. It is a way to allow losses to play a part in the optimization problem.

$$L_i = \frac{1}{1 - \frac{\partial P_L}{\partial P_{Gi}}}$$

change in losses $\rightarrow \frac{\partial P_L}{\partial P_{Gi}}$ change in the real power of generator i.

IF $L_i > 1$: penalized
 $L_i < 1$: incentivized.

(c) What is a security constant in an optimal power flow problem?

The dispatch at any point in time should generally have contingency (e.g. N-1) so that the power system remains robust

For each contingency event, adding a set of constraints for this to ensure feasibility results in security-constrained OPF (SCOPF).

Usually only a few contingencies are checked, and used in a DC OPF to save time.

- (d) An island power company with a network comprising 10 controllable generators, 100 buses and 100 branches wants to run a security-constrained AC optimal power flow. If there are five contingencies which must be fully included, estimate the size of the problem. The size of the problem is defined by the number of design variables and number of constraints. There are no other design variables except those associated with the generators (i.e. any regulators, tap settings, etc. are not included in the OPF problem).

Design variables: P, V at generators ($= 20$)
(or P, Q)
(or just P for some)

Constraints: generator max/min (20×2)

Bus power balance: 100×2 (200)
 $\hookrightarrow P \text{ \& } Q$

Branch-flows: 100×1 (100)
~~max limit~~

Bus voltage limits: 100×2 (200)
 $\hookrightarrow \text{min/max}$

10-20	design variables
500+	constraints

Note 1: DC OPF would have only:
 $\rightarrow$ branch-flow (100)
 $\rightarrow$ Power-balance (P only)
 ~ 200 ^{network} constraints, all linear

Note 2: Exact numbers don't matter.
 Show good reasoning.

- (e) For the same scenario as the previous question, estimate the size of the economic dispatch, DC OPF, and AC-OPF (no security constraints) problems. In addition to scaling / size issues, what makes the the AC-OPF and SC-ACOPF difficult to solve?

As before

DC OPF 10 design variables
20 generator limits + 200 bus/branch constraints.

Economic dispatch: 10 design variables
20 min/max generator limits
+ 1 overall power balance.

→ Non-linear constraints / non-convexity of the problem makes ACOPF and SC-ACOPF much harder.

- (f) Consider the three-bus system depicted in Fig. 1. You may assume that each meter is very close to the nearest bus in the figure.

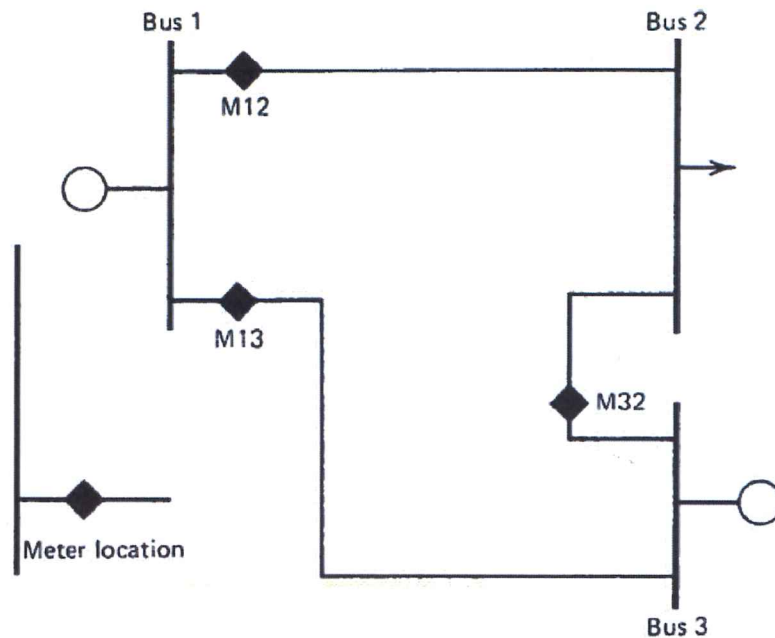


Figure 1: Power system for question 4 (e)

- a) Using the DC power-flow approximation derived in lectures, solve for the angles of the network if bus 1 (not bus 3 as in the lecture notes) is taken as the reference. The network impedances are $X_{12} = 0.2$, $X_{13} = 0.4$, $X_{23} = 0.25$ and the measurements were $M_{12} = 62$ MW, $M_{13} = 6$ MW, $M_{32} = 37$ MW. Assume all meters are equivalent.

$$\text{Use } x^* = [H^T R^{-1} H]^{-1} H^T R^{-1} z_{\text{meas}} = \begin{bmatrix} \theta_2^* \\ \theta_3^* \end{bmatrix}$$

~~$$x^* = [H]^T [H H^T]^{-1} z_{\text{meas}}$$~~

$$H : M_{12} = \frac{1}{0.2} (\theta_1 - \theta_2) = -5\theta_2$$

$$M_{13} = \frac{1}{0.4} (\theta_1 - \theta_3) = -2.5\theta_3$$

$$M_{32} = \frac{1}{0.25} (\theta_3 - \theta_2) = 4\theta_3 - 4\theta_2$$

$$[H] = \begin{bmatrix} -5 & 0 \\ 0 & -2.5 \\ -4 & 4 \end{bmatrix}$$

$$x^* = \begin{bmatrix} -0.123 \\ -0.029 \end{bmatrix}$$

exam won't involve much computation.

TURN OVER

→ recommend solving on computer.

b) Taking bus 3 as a reference (as in the lecture example), suppose the measurement M_{13} is taken by a better meter with ± 0.3 MW accuracy. With the aid of a computer if necessary, calculate the weighted least squares estimate of the voltage angles and buses 1 and 2.

In this case

$$H = \begin{bmatrix} 5 & -5 \\ 2.5 & 0 \\ 0 & -4 \end{bmatrix}$$

$$R = \begin{bmatrix} 0.0001 & & \\ & 10^{-6} & \\ & & 10^{-4} \end{bmatrix}$$

$$x^k = \begin{bmatrix} 0.0241 \\ -0.0970 \end{bmatrix} = \begin{bmatrix} \theta_1^* \\ \theta_2^* \end{bmatrix}$$

was 0.0286 before.
was 0.0943

c) If the DC power-flow approximation is **NOT** used, and **all bus voltages, along with reactive power at the meter locations, are measured** as well, formulate (do NOT solve) an appropriate non-linear least squares problem. Assume all measurements (including those of the bus voltages) have the same standard deviation in p.u. and thus weightings can be ignored.

The active power flow through the line from i to j is given by:

$$P_{ij} = V_i V_j B_{ij} \sin(\theta_i - \theta_j).$$

The reactive power leaving bus i through the line from i to j is given by:

$$Q_{ij} = V_i (V_i - V_j \cos(\theta_i - \theta_j)) B_{ij}.$$

$$\min_{V_1, V_2, V_3, \theta_1, \theta_2} \left\{ \begin{aligned} & \sum_{(i,j) \in M} \left(P_{ij}^m - V_i V_j B_{ij} \sin(\theta_i - \theta_j) \right)^2 \\ & + \sum_{(i,j) \in M} \left(Q_{ij}^m - V_i (V_i - V_j \cos(\theta_i - \theta_j)) B_{ij} \right)^2 \\ & + \sum_{k \in \{1,2,3\}} \left(V_i^m - V_i \right)^2 \end{aligned} \right\} \quad \begin{array}{l} f(x) \\ \uparrow \\ \text{objective function} \end{array}$$

measured (pointing to P_{ij}^m)

all measured lines (pointing to M)

$M := \{(1,2), (1,3), (3,2)\}$

$\theta_3 = 0 \leftarrow \text{reference bus (any choice is fine)}$

No need to solve !!